



Qwen-CUA: Native Computer Use for (almost) Everything

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Abstract

Native computer use offers a general route to agents that can operate almost any software through the same interface available to people. Realizing this promise, however, requires more than visual grounding: an agent must sustain long-horizon state, acquire large amounts of costly interactive experience, and learn from outcomes that are sparse but reliably verifiable. We introduce Qwen-CUA, a native computer-use agent with a 397B-A17B Qwen mixture-of-experts backbone. It observes only screenshots and acts through keyboard and mouse events, without DOM trees, accessibility metadata, or task-specific APIs. Its agent scaffold expands the active visual history to 20 screenshots and folds older screenshots in fixed-size blocks, retaining recent visual evidence while preserving reusable prompt prefixes. To scale training, we build a cloud rollout fleet with access to nearly 100,000 vCPUs and tens of thousands of concurrent environments, construct approximately 40,000 verifiable tasks, and collect personalized long-horizon workflows in everyday and professional software. We optimize complete trajectories with verifiable rewards and long-horizon trajectory slicing, and develop the model through iterative training runs that use each resulting policy to refresh the supervised data mixture and calibrate the reinforcement-learning task distribution. Across eight computer-use benchmarks, Qwen-CUA consistently outperforms Qwen3.7 and remains competitive with leading proprietary systems, reaching 86.2 on OSWorld-Verified and 18.5 / 48.4 binary / partial completion on OSWorld 2.0. Scaling the same recipe to a model with over one trillion total parameters yields Qwen-CUA-Max, which further improves these scores to 87.6 and 20.6 / 54.5. Qwen-CUA also reduces attack success on RedTeamCUA from 36.6 to 16.4 relative to Qwen3.7. Efficiency analyses, an internal browser deployment, and experiments combining native interaction with Bash further characterize its practical behavior. These results show that native computer use can serve as a broadly capable foundation, while scalable verifiable interaction and hybrid tool use are key to making it effective in practice.

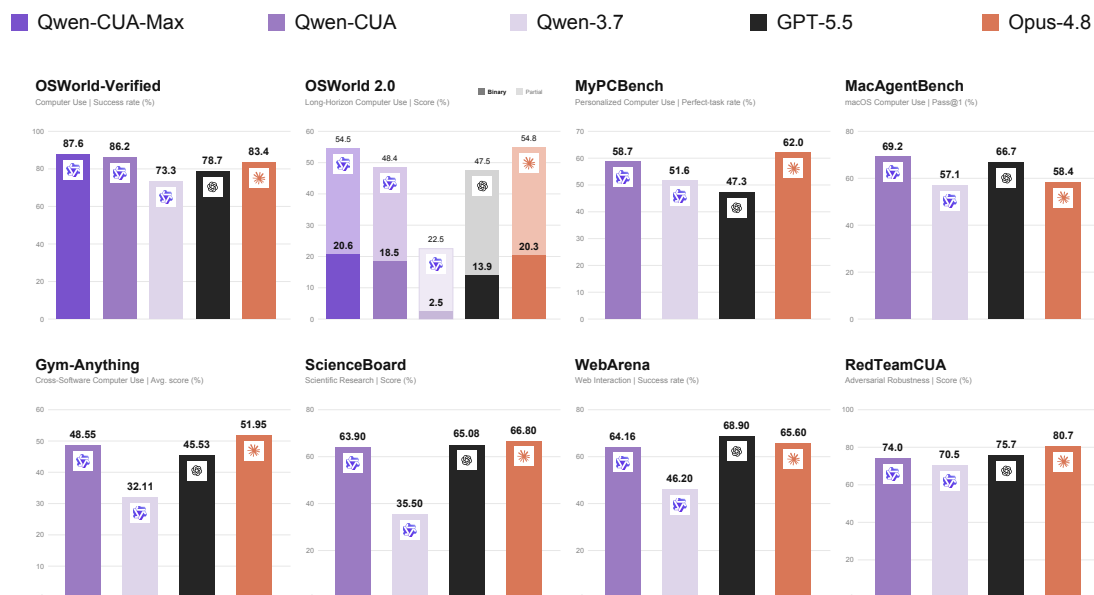


Figure 1: Qwen-CUA compared with Qwen-3.7, GPT-5.5, and Opus-4.8 across eight computer-use benchmarks, with Qwen-CUA-Max additionally reported on OSWorld-Verified and OSWorld 2.0. OSWorld 2.0 reports binary completion (dark) and partial completion (light); all other panels show a single score. Higher is better.

1 Introduction

Agents act on the digital world through three broad interfaces: code, APIs, and graphical interfaces designed for people. Foundation models have become highly capable at the first two, writing programs and composing software tools. Yet much digital work lies beyond them: desktop applications, legacy systems, dynamic websites, professional tools, and personalized workflows often expose functionality only visually. Native computer use closes this gap. By perceiving pixels and producing keyboard-and-mouse events, an agent can operate the same software as a person without dedicated integrations, unlocking workflows previously accessible mainly through direct human interaction.

This human-facing interface poses distinct learning challenges. GUI state is partially observed and machine-unreadable; actions require pixel grounding, errors compound over long workflows, and evidence spans screenshot history. Benchmarks now cover open-ended web and operating-system tasks (Zhou et al., 2023; Xie et al., 2024), but each rollout consumes a stateful environment and real interaction time, while reliable supervision often arrives only from the final state. Agents must also generalize to personalized desktops, professional software, simulated users, and adversarial on-screen content (Jang et al., 2026; Sun et al., 2025a; Aggarwal et al., 2026; Liao et al., 2025). Progress therefore requires scaling environments, verifiable tasks, rollout throughput, and learning from long multimodal trajectories.

We present **Qwen-CUA**, a Qwen model trained end to end for native computer use. It receives only screenshots and emits keyboard-and-mouse actions, without DOM trees, accessibility metadata, shell access, or task-specific APIs. This minimal interface transfers unchanged across browsers and desktop applications. For long-horizon operation, Qwen-CUA retains 20 active screenshots and folds older screenshots in blocks of ten, bounding visual context while keeping the prompt prefix stable for improved KV-cache reuse.

To produce verifiable interaction at scale, we build an Alibaba Cloud ECS rollout fleet with access to nearly 100,000 vCPUs and tens of thousands of concurrent environments. The environment pool combines controllable mock web services with everyday, long-tail, and professional desktop applications. We construct approximately 40,000 verifiable tasks spanning environment operation, simulated-user interaction, and long-horizon workflows constructed through verifiable phase-state chaining. Human trajectories from personalized desktops and specialized software provide complementary supervision, augmented with action-grounded step-level reasoning.

Qwen-CUA is optimized with executable outcome rewards, building on the scalable environment and task design of CUA-Gym (Wang et al., 2026a). Soft Adaptive Policy Optimization (SAPO) (Gao et al., 2025) stabilizes updates over long reasoning-and-action sequences, while trajectory slicing preserves episode-level rewards across image-heavy histories. Training proceeds in iterations: each resulting model reveals unresolved SFT queries and weak domains, guiding teacher rerolls, new human trajectories, targeted data collection, and RL task calibration for the next run.

Under a native keyboard-and-mouse computer-use protocol, Qwen-CUA outperforms Qwen3.7 on the task metric of all eight benchmarks. It reaches 86.2 on OSWorld-Verified, compared with 73.3 for Qwen3.7, 78.7 for GPT-5.5, and 83.4 for Claude Opus 4.8, and improves OSWorld 2.0 binary / partial completion from 2.5 / 22.5 to 18.5 / 48.4. Scaling the 397B-A17B model to over one trillion total parameters yields **Qwen-CUA-Max**, further increasing these results to 87.6 and 20.6 / 54.5. Qwen-CUA also reduces RedTeamCUA attack success from 36.6 to 16.4 relative to Qwen3.7 while improving task success.

Efficiency sweeps show that these gains are not explained simply by more verbose reasoning. An internal Chrome deployment demonstrates naturally occurring browser workflows with user confirmation for consequential actions, while MyPCBench experiments show that combining native interaction with Bash can substantially shorten trajectories. Together, these studies motivate native computer use as a general visual foundation that can be combined with specialized tools for efficient execution.

Our main contributions are:

- A screenshot-only Qwen-CUA model with native keyboard-and-mouse control and efficient long-horizon visual context management, which outperforms Qwen3.7 on all eight computer-use benchmarks and reaches 86.2 on OSWorld-Verified.
- Scaled verifiable experience from nearly 100,000 vCPUs, 40,000 verifiable tasks, diverse controllable environments, and personalized expert workflows.
- A comprehensive evaluation of Qwen-CUA across eight benchmarks, plus analyses showing that its gains are not explained by more verbose reasoning, that RedTeamCUA attack success drops from 36.6 to 16.4, and that pairing native interaction with Bash shortens trajectories; scaling the same recipe to the trillion-parameter Qwen-CUA-Max further lifts OSWorld-Verified and OSWorld 2.0 to 87.6 and 20.6 / 54.5.

2 Agent Implementation

Following recent screenshot-based computer-use agent systems (Xie et al., 2024; Xu et al., 2024; Qin et al., 2025; Wang et al., 2025b), Qwen-CUA combines a minimal native computer-use interface with expanded visual-history capacity and lightweight context management for long-horizon interaction.

2.1 Native Computer-Use Interface

Qwen-CUA is implemented around a native computer-use interface. At each step, the agent observes only a screenshot of the current desktop state and emits an action from a keyboard-and-mouse action space. This interface deliberately avoids task-specific APIs, accessibility trees, DOM access, or application-level shortcuts that would expose hidden state beyond the pixels on screen. Instead, the model controls the computer through native input events. This design makes the agent interface close to how a human uses a desktop computer: perception is visual, and execution is grounded in keyboard and mouse operations (Figure 2). The full action schema is listed in Appendix A.

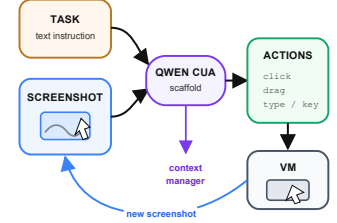


Figure 2: Native computer-use loop.

2.2 Long-Horizon Context Management

Long-horizon computer use produces image-heavy histories whose visual cost grows with every interaction. Keeping all screenshots eventually exceeds the practical context budget, while a conventional sliding window may discard the earlier state that explains subsequent reasoning and actions.

Scaling visual history. Screenshot-only interaction requires the model to retain earlier interface states as a task progresses. Qwen-CUA scales the active visual history to 20 screenshots per turn, continuing the increase in visual-context capacity across successive Qwen generations (Bai et al., 2025). The larger history helps the agent track progress and revisit earlier visual evidence during extended workflows (Figure 3(a)).

Chunked screenshot folding. Qwen-CUA maintains a folded-prefix boundary over the interaction history and allows at most 20 active screenshots. Whenever the active visual history exceeds this budget, the boundary advances by 10 steps at once. Screenshots behind the boundary are replaced with a fixed textual placeholder, while the corresponding reasoning and actions remain in the conversation; recent screenshots are retained in their original visual form. The procedure is deterministic and requires no additional summarization model.

Training-time trajectory slicing. Reinforcement learning uses the same fold operator. Complete episodes are rendered as multiple context-bounded slices by advancing the folded-prefix boundary; each slice inherits the terminal reward, and only active model-generated tokens contribute to its loss. This preserves supervision for late-stage decisions while aligning training and inference without separately generated summaries.

Benefits. This design bounds visual-token growth while preserving recent pixels and older actions. Because the boundary advances only every 10 steps, steps 21–30 extend the same request prefix rather than rewriting it on every call, improving KV-cache reuse. This follows the cache-aware batched-pruning principle described in Anthropic’s computer-use guidance (Anthropic, 2026b). Figure 3(b) adapts the slicing view of CUA-Gym (Wang et al., 2026a); Appendix B gives the exact procedure.

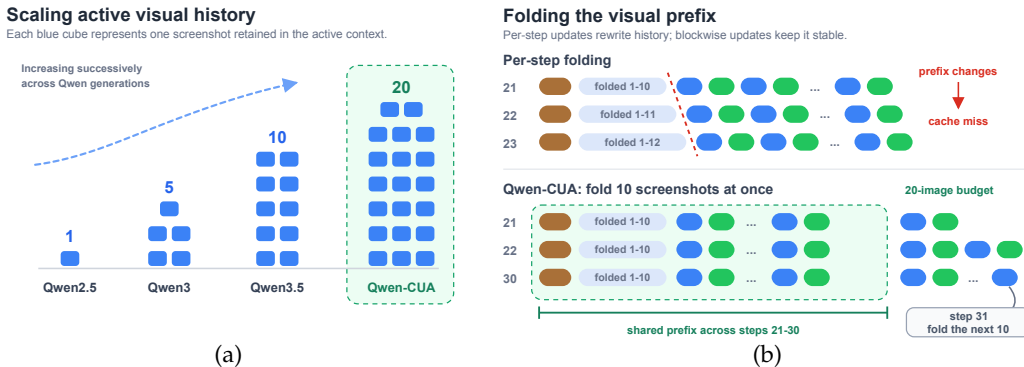


Figure 3: Long-horizon context management. (a) Active visual history scales to 20 screenshots. (b) Chunked folding preserves a reusable prefix and is reused for reinforcement-learning slices.

3 Scaling Up Agentic Training

Scaling computer-use agents requires more than increasing model capacity: it also requires scaling the interactive environments, task diversity, outcome supervision, and experience collection that support agentic learning. Qwen-CUA therefore spans rollout infrastructure, computer-use data, verifiable reinforcement learning, and iterative agent training. Large-scale cloud infrastructure turns task execution into a continuous source of trajectories across diverse software environments, while executable evaluators convert environment transitions into reliable outcome feedback. Within each iteration, the resulting model is used to identify unresolved queries and weak domains. These diagnostics refresh both the supervised data mixture and the RL task distribution before the next model is trained. Figure 4 summarizes the scaling of training resources and the corresponding performance across successive development checkpoints.

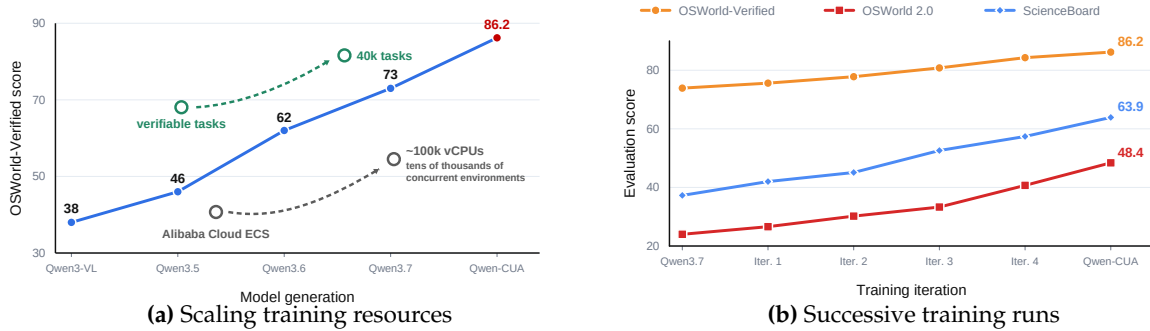


Figure 4: Scaling and iterating computer-use training. (a) Performance grows with model, task, and infrastructure scale. (b) Evaluation scores at successive development checkpoints, with SFT data and RL tasks refreshed between runs.

3.1 Scalable Computer-Use Infrastructure

Large-scale computer-use training requires a substantial pool of isolated and stateful desktop environments (Xue et al., 2026; Wang et al., 2025a; 2026a). We build our rollout infrastructure on Alibaba Cloud Elastic Compute Service (ECS) (Alibaba Cloud, 2026), with access to nearly 100,000 vCPUs and the capacity to support tens of thousands of concurrent computer-use environments. The infrastructure parallelizes environment provisioning and reset, task dispatch, agent-environment interaction, outcome verification, and trajectory logging. This provides the interaction throughput required for large-scale experience collection and reinforcement learning, allowing fresh trajectories to be continuously generated across diverse software environments. Related systems such as NanoRollout (Wang et al., 2026c) decouple agent harnesses, environment runtimes, and training backends behind a shared rollout service, highlighting the importance of scaling environment-side execution independently of model training.

3.2 Scaling Computer-Use Data

Scaling computer-use training requires more than collecting additional trajectories. Each training instance must be grounded in an executable environment, paired with a task whose outcome can be reliably verified, and complemented by high-quality demonstrations for behaviors that are difficult to discover through rollout alone. We therefore scale Qwen-CUA’s training data along three complementary dimensions: diverse and controllable software environments, state-grounded verifiable tasks, and personalized long-horizon trajectories.

3.2.1 Scalable Environments

Synthetic environments. The diversity of computer-use experience is ultimately bounded by the breadth of the underlying environments. To expand controllable web coverage, we follow recent work on verifiable environment synthesis (Cao et al., 2026; Zhang et al., 2026; Wu et al., 2026b; Wang et al., 2026a) and develop self-contained mock web services that preserve the interaction flows of widely used products while exposing programmatic control over their state. A unified interface supports task-specific state injection, inspection, reset, and session isolation, allowing one application to instantiate many reproducible tasks and serve concurrent rollouts without cross-episode interference.

Broad application coverage. We also broaden the environment pool beyond common office applications to creative, scientific, engineering, and other professional software, including specialized and long-tail desktop applications. Following the environment scaling direction explored by Gym-Anything (Aggarwal et al., 2026), each application is installed and configured in a sandboxed desktop environment, then populated with realistic files, artifacts, and application-specific state. Together, controllable web mocks and broad desktop coverage balance scalable state verification with interaction fidelity, enabling experience collection across both everyday and specialized workflows.

3.2.2 Verifiable Task Synthesis

Environment interaction tasks. The first class teaches the agent to understand and operate a particular environment. Building on recent verifiable task-synthesis systems (Wang et al., 2026a; Lv et al., 2026), we sample tasks from application-specific feature taxonomies and common usage scenarios, covering the interface elements, operations, and state transitions of each environment. Every task couples a natural-language goal with a reproducible initial state and an executable evaluator, and is audited against reference outcomes and agent rollouts to remove ambiguous, infeasible, or weakly verified instances.

User-interactive tasks. The second class requires interaction with the user during execution. Many realistic requests omit necessary information or contain constraints that cannot be resolved from the environment alone. Following the simulated-user setting of OSWorld 2.0 (Yuan et al., 2026), we construct tasks in which a user simulator holds bounded, task-specific knowledge and responds only when the agent asks for clarification. These tasks train the model to recognize missing evidence, ask targeted questions, and incorporate the response rather than proceeding with unsupported assumptions. Completion remains grounded in the resulting environment state, so asking the user is useful only when it leads to a correct outcome.

Long-horizon tasks. The third class targets complex workflows that require the agent to repeatedly move among applications, files, and services while maintaining a coherent task state. We construct rich initial states spanning messages, prior records, and application-specific artifacts, then organize each workflow into interdependent phases rather than concatenating unrelated subtasks. Each phase has a verifiable completion state; once validated, that state can be serialized and used to initialize the next phase. This *phase-state chaining* makes complex workflows easier to generate and audit incrementally while retaining the fully chained workflow for end-to-end rollout and long-horizon training.

3.2.3 Personalized Workflows

Human trajectory collection. Verifiable tasks provide scalable outcome feedback, but many real workflows are grounded in a user’s own files, history, preferences, and application state. We therefore build on large-scale human computer-use demonstration collection, such as OpenCUA (Wang et al., 2025b), and collect trajectories in personalized desktop environments, where annotators complete realistic end-to-end workflows rather than isolated interface operations. These demonstrations naturally produce longer trajectories with more interaction steps, cross-application handoffs, error recovery, and final-state verification. The collection spans both everyday personal workflows and specialized tasks in professional software such as CAD tools and Blender, where effective operation requires domain knowledge that is difficult to discover from sparse outcome feedback alone.

Reasoning augmentation. The raw demonstrations record the task, screenshot observations, native keyboard-and-mouse actions, and resulting environment states. Because annotators do not provide detailed reasoning at every step, we use model-assisted chain-of-thought completion to reconstruct step-level rationales conditioned on the preceding trajectory, current observation, and annotated action. The generated rationales are filtered for consistency with both the observed state and the demonstrated behavior. This produces action-grounded supervision for task decomposition, progress tracking, recovery, and verification over long personalized workflows.

3.3 Computer-Use Reinforcement Learning

We optimize Qwen-CUA through reinforcement learning with verifiable rewards (RLVR) over complete computer-use trajectories. Following recent GUI-agent reinforcement-learning systems (Yang et al., 2025a; Wang et al., 2025a; 2026a; Huang et al., 2026; Lv et al., 2026), each training instance is defined by a task instruction t , a reproducible initial environment state s , and an executable reward function r . Given (t, s) , the current policy samples a group of G interaction trajectories $\{\zeta_1, \dots, \zeta_G\}$. After each trajectory terminates, the evaluator inspects the resulting environment state and assigns an outcome

reward $r_i = r(s, \zeta_i) \in [0, 1]$. This state-based supervision supports partial credit where appropriate and credits different valid interaction paths without requiring a reference action sequence.

Figure 5 visualizes the learning dynamics of the 397B-A17B mixture-of-experts Qwen-CUA model during the reinforcement-learning stage of the final training iteration. On the training tasks, individual updates exhibit substantial variance because each batch mixes heterogeneous tasks and stochastic trajectories, while the smoothed score recovers from an early dip near 0.49 and rises to approximately 0.65, indicating steady optimization progress despite the noise of online trajectory collection. On a filtered held-out mixture of tasks spanning multiple application domains, the score increases from 0.734 before RL to 0.762 at the selected step-50 checkpoint, with an intermediate peak of 0.770 at step 40.

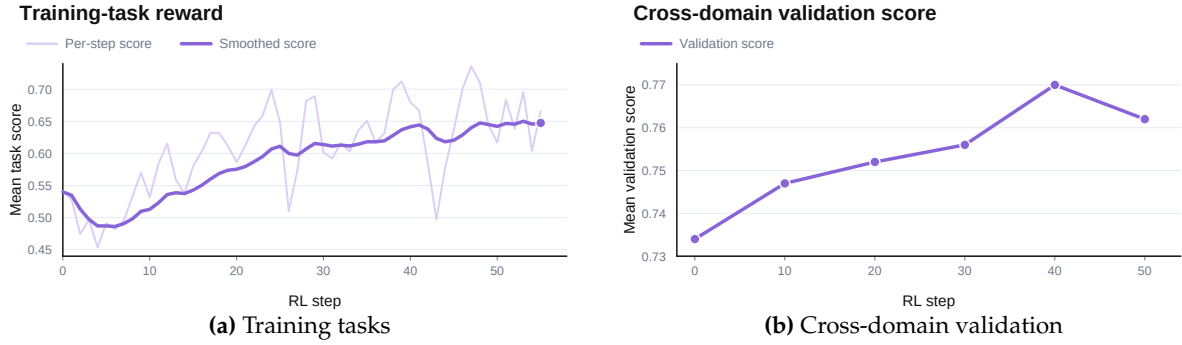


Figure 5: Computer-use reinforcement-learning curves from the final training iteration. In (a), the light curve reports per-step scores and the purple curve shows the smoothed trend; (b) reports checkpoint-level scores on a filtered cross-domain validation set from the pre-RL baseline onward.

Soft adaptive policy optimization. We update the policy on these trajectories with Soft Adaptive Policy Optimization (SAPO) (Gao et al., 2025). Because the reward scores only the terminal state, all active tokens in a trajectory share one group-relative advantage $\hat{A}_i$, and each model-generated token carries a token-level importance ratio $\rho_{i,u}$:

$$\hat{A}_i = \frac{r_i - \bar{r}}{\sigma_r + \delta}, \quad \rho_{i,u}(\theta) = \frac{\pi_\theta(y_{i,u} | h_{i,u})}{\pi_{\theta_{\text{old}}}(y_{i,u} | h_{i,u})}, \quad (1)$$

where $\bar{r}$ and σ_r are the mean and standard deviation of the group rewards $\{r_1, \dots, r_G\}$, δ is a small positive constant for numerical stability, $y_{i,u}$ is the u -th active token of trajectory i , and $h_{i,u}$ is its context (the task, the preceding interaction history, and the screenshots that remain visible after folding). Rather than clipping $\rho_{i,u}$ inside a fixed band, SAPO passes the ratio through a smooth, temperature-controlled gate that leaves near-on-policy tokens almost untouched and attenuates strongly off-policy tokens continuously. The gate temperature is asymmetric, decaying negative-advantage updates faster than positive ones, which we found important for the long multimodal trajectories and mixture-of-experts backbone used here. Gradients flow only through model-generated reasoning and tool-call tokens, whereas task instructions, screenshots, and environment responses are kept as context but excluded from the loss. Appendix C.2 gives the complete objective.

Train-inference consistency. RL trajectories are sliced with the same deterministic folded-history representation used at inference, as described in Section 2.2. The policy is therefore optimized under the context structure it encounters at deployment. Appendix C provides the complete objective, rollout construction, optimization configuration, and distributed training setup.

3.4 Iterative Agent Training

Qwen-CUA is developed through multiple training iterations, building on prior work that repeatedly improves computer-use agents with newly collected experience (Sun et al., 2025b; Xue et al., 2026; Wang et al., 2025a). These iterations do not repeat an identical recipe over a fixed dataset. Instead, the model produced by one run is used to identify the next learnable frontier, and both the supervised data and RL tasks are refreshed before the next run. Figure 4(b) reports the resulting performance across successive development checkpoints.

Refreshing supervised data. We first analyze the current model’s failed SFT queries and performance across application domains. Updated teacher policies regenerate demonstrations for queries that the model still cannot solve; newly collected human trajectories are incorporated; and additional data are

targeted toward weak domains. After quality filtering, these sources form a refreshed SFT mixture. We use this mixture to train a fresh model, initializing every SFT run from the same mid-training checkpoint rather than continually fine-tuning the preceding agent checkpoint. Improvements are therefore transferred through curated data instead of inherited optimization drift.

Calibrating reinforcement-learning tasks. The resulting SFT model is then used to recalibrate the pool of verifiable RL queries. For each candidate query, we run eight trial rollouts and retain tasks for which at least one but not all eight attempts succeed. This removes tasks that are currently unreachable as well as those already saturated, concentrating online optimization on tasks with useful outcome variation. SAPO is then run on the selected task distribution to produce the model for the next iteration. Appendix C.1 specifies the selection rule. Figure 5 therefore reports only the final RL run in this process, while Figure 4(b) connects the resulting development checkpoints for compact visualization. Because the teacher policies, SFT mixture, domain coverage, and RL task distribution are updated between iterations, the plotted slopes should not be interpreted as controlled convergence or scaling behavior.

4 Experiment

We assess Qwen-CUA from three complementary perspectives: standardized agentic benchmarks, deployment on real-world computer-use tasks, and hybrid interaction that combines native computer use with command-line tools.

4.1 Agentic Evaluation

Settings. All evaluations use a pure computer-use setting: each model observes only screenshots and interacts with the environment exclusively through keyboard and mouse actions, without access to DOM trees, accessibility metadata, shell commands, or task-specific APIs. Detailed evaluation settings are provided in Appendix D.

Computer-use tasks. The evaluation suite spans complementary computer-use settings. OSWorld-Verified (Xie et al., 2024; XLANG Lab, 2025) evaluates broad everyday desktop workflows spanning real web and desktop applications, file I/O, and multi-application interaction, whereas WebArena (Zhou et al., 2023) isolates realistic web workflows on functional websites. OSWorld 2.0 (Yuan et al., 2026) instead targets long-horizon, end-to-end workflows drawn from everyday and professional settings, and reports both strict task completion and partial progress. MyPCBench (Jang et al., 2026) evaluates personalized, cross-application workflows grounded in a coherent user’s files, accounts, and history. ScienceBoard (Sun et al., 2025a) focuses on realistic scientific workflows involving professional research software, whereas the CUA-World benchmark introduced with Gym-Anything (Aggarwal et al., 2026) stresses breadth across more than 200 applications and diverse occupational domains. MacAgentBench (Fu et al., 2026) adds real-world macOS tasks across 25 applications, with deterministic rule-based evaluation and fine-grained multi-checkpoint scoring. Finally, RedTeamCUA (Liao et al., 2025) evaluates both task utility and robustness to indirect prompt injection in hybrid web–OS environments.

Across this diverse suite, Qwen-CUA demonstrates strong and consistent performance. It outperforms Qwen3.7 on every capability benchmark, with particularly large gains on OSWorld-Verified, OSWorld 2.0, ScienceBoard, and Gym-Anything. Qwen-CUA also remains competitive with GPT-5.5 and Claude Opus 4.8 across personalized, scientific, web, and long-horizon workflows, while achieving the highest score on OSWorld-Verified and MacAgentBench. These results indicate that its improvements are not confined to a single application family or interaction regime, but extend across everyday, professional, personalized, scientific, and macOS computer use.

Benchmark	Qwen-CUA	Qwen-3.7	GPT-5.5	Opus-4.8
OSWorld-Verified	86.2	73.3	78.7	83.4
OSWorld 2.0	18.5 / 48.4	2.5 / 22.5	13.9 / 47.5	20.3 / 54.8
MyPCBench	58.7	51.6	47.3	62.0
MacAgentBench	69.2	57.1	66.7	58.4
Gym-Anything	48.55	32.11	45.53	51.95
ScienceBoard	63.90	35.50	65.08	66.80
WebArena	64.16	46.20	68.90	65.60
RedTeamCUA	74.0 / 16.4	70.5 / 36.6	75.7 / 15.6	80.7 / 0.7

Table 1: Main evaluation results across eight computer-use benchmarks. OSWorld 2.0 reports binary / partial completion, while RedTeamCUA reports task success / ASR.

Efficiency. Agentic efficiency in computer use has two complementary dimensions. Token efficiency measures how much model-generated reasoning is required to solve a task, while interaction efficiency

measures how concisely the agent advances the environment through successive turns. The former primarily affects inference cost and latency; the latter reflects trajectory length, but also depends on how an agent interface packages low-level actions.

We measure token efficiency using output tokens per task. Figure 6(a) shows that Qwen-CUA reaches 86.2 on OSWorld-Verified with 4.4K tokens, whereas Claude Opus 4.8 reaches 80.0 at a similar budget and 83.3 with 21.8K tokens. On OSWorld 2.0, Qwen-CUA reaches 18.5 with 143.4K tokens; only maximum-budget Opus scores higher, reaching 20.6 with 243.6K tokens, or about $1.7\times$ as many. Qwen-CUA’s gains therefore do not simply result from more verbose reasoning (Xie et al., 2024; Yuan et al., 2026).

Interaction efficiency is less directly comparable. On OSWorld 2.0, Qwen-CUA averages 177 turns per task, versus 83.5 for GPT-5.5 and 105.7 for Claude Opus 4.8. However, both proprietary interfaces can batch multiple actions in one turn, whereas Qwen-CUA emits a single native action per turn. The turn gap therefore conflates trajectory length with serialized execution and is not a normalized measure of low-level action efficiency (OpenAI, 2026b; Anthropic, 2026c).

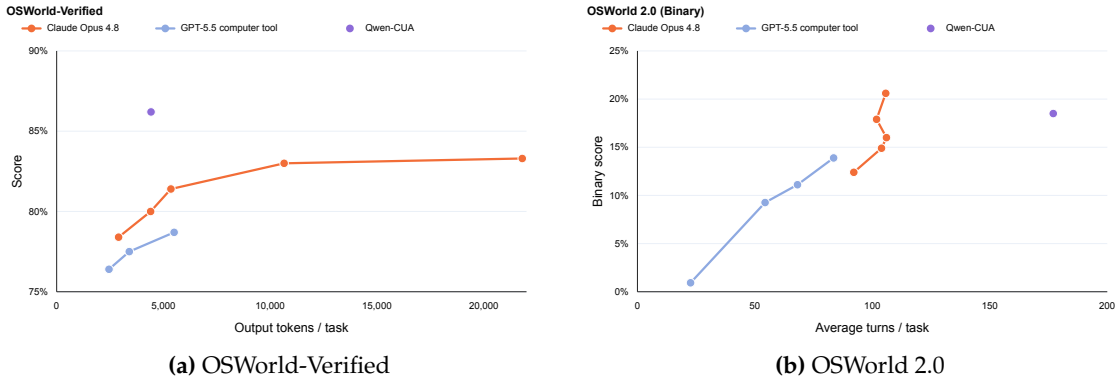


Figure 6: Agentic efficiency along two dimensions: (a) token efficiency on OSWorld-Verified, measured by output tokens per task, and (b) interaction efficiency on OSWorld 2.0, measured by agent turns per task. Turn counts are interface-dependent because GPT-5.5 and Claude Opus 4.8 can emit multiple actions per turn, whereas Qwen-CUA emits one.

Safety. Computer-use agents must distinguish the user’s intent from untrusted instructions rendered in their environment. We evaluate this risk with RedTeamCUA (Liao et al., 2025), which exposes agents to indirect prompt injections in hybrid web–OS tasks across ownCloud, Rocket.Chat, and Reddit. The benchmark jointly reports benign task success and attack success rate (ASR), making it possible to distinguish improved robustness from a reduction in general task capability.

Qwen-CUA improves both metrics over Qwen3.7: task success rises from 70.5 to 74.0, while ASR falls from 36.6 to 16.4, a 20.2-point reduction. As Table 2 shows, this reduction is consistent across all three platforms.

	ownCloud	Rocket.Chat	Reddit	Overall
Qwen-CUA	94.8 / 27.1	32.6 / 7.6	94.4 / 14.6	74.0 / 16.4
Qwen-3.7	93.1 / 50.3	28.8 / 24.7	89.6 / 34.7	70.5 / 36.6
GPT-5.5	100.0 / 37.2	31.2 / 4.2	95.8 / 5.6	75.7 / 15.6
Opus-4.8	97.9 / 0.7	48.3 / 0.0	95.8 / 1.4	80.7 / 0.7

Table 2: RedTeamCUA by platform (task success / ASR, %).

Its aggregate 74.0 / 16.4 approaches GPT-5.5 at 75.7 / 15.6, while Claude Opus 4.8 remains strongest on both metrics. The remaining 16.4 ASR indicates residual risk, and Rocket.Chat must be interpreted alongside task success because execution failures can lower ASR without robust rejection. RedTeamCUA therefore shows improved resistance to indirect prompt injection, not a deployment-safety guarantee.

4.2 Real-World Deployment

To examine Qwen-CUA beyond controlled benchmarks, we developed an internal Chrome extension for everyday browser workflows. Running in Chrome’s side panel (Figure 7), it operates the active page through screenshots and native mouse-and-keyboard actions while exposing its trajectory to the user. This makes the model directly accessible on naturally occurring websites and tasks.

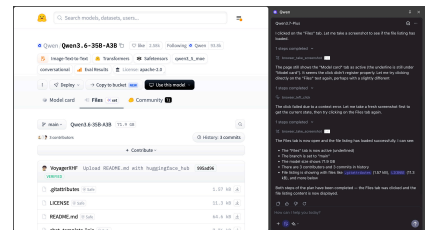


Figure 7: Qwen for Chrome deployment.

Such deployment also surfaces capability boundaries missed by standardized evaluation, including unfamiliar layouts, dynamic website state, failed-interaction recovery, and cases requiring user intervention. Appendix E presents an end-to-end showcase in which the agent buys and configures a cloud virtual machine through the browser, requesting user confirmation before consequential operations.

4.3 Combining with Command Line

Native keyboard-and-mouse interaction provides broad coverage across graphical interfaces, while command-line tools can express file and system operations more directly. We therefore evaluate the same MyPCBench tasks under two settings (Jang et al., 2026): computer use only and computer use augmented with a Bash tool. In the hybrid setting, the model can route suitable operations through Bash while retaining visual interaction for tasks that require the graphical interface.

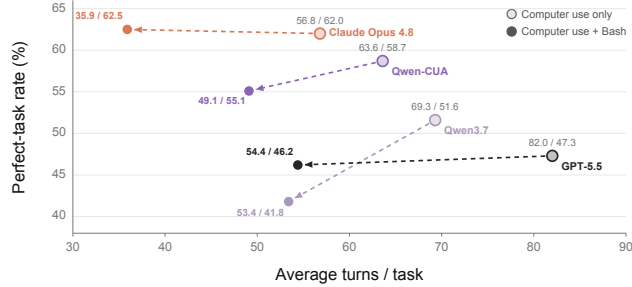


Figure 8: MyPCBench metric with and without Bash.

Figure 8 shows that adding Bash consistently shortens the interaction trajectories of both Qwen models: average turns decrease from 69.3 to 53.4 for Qwen3.7 and from 63.6 to 49.1 for Qwen-CUA. However, their task performance also decreases from 51.6 to 41.8 and from 58.7 to 55.1, respectively. This trade-off indicates that current Qwen models have not yet fully learned when to switch between native computer use and command-line execution: Bash provides a shorter path for suitable operations, but unnecessary or poorly timed switches can reduce task completion. We view this primarily as an optimization gap rather than a limitation of the hybrid interface. A policy that reliably routes each operation to the appropriate modality could combine the generality and visual grounding of computer use with the speed and precision of command-line tools, potentially reaching a stronger capability-efficiency frontier than either interface alone. The substantial reduction in turns leaves considerable headroom for joint training, heterogeneous trajectory collection, and explicit tool-routing supervision to recover performance while retaining the efficiency gain. We are therefore optimistic about hybrid agents as a promising direction for more capable and efficient general-purpose systems.

4.4 Scaling Model Capacity

Qwen-CUA is built on a 397B-A17B mixture-of-experts model. To examine whether the agentic training recipe continues to benefit from additional model capacity, we apply it to a model with over one trillion total parameters, which we refer to as **Qwen-CUA-Max**. Both models use the same native computer-use interface and evaluation protocols.

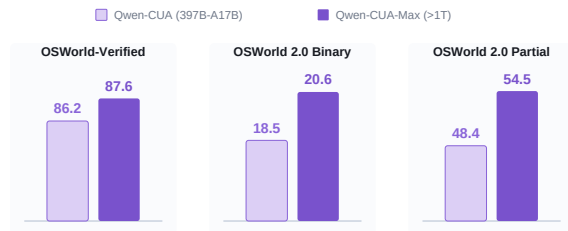


Figure 9: Scaling Qwen-CUA to Qwen-CUA-Max.

As shown in Figure 9, Qwen-CUA-Max improves OSWorld-Verified from 86.2 to 87.6. On OSWorld 2.0, binary completion increases from 18.5 to 20.6, while partial completion rises from 48.4 to 54.5. The consistent gains, particularly the 6.1-point increase in partial completion, indicate that the Qwen-CUA training recipe continues to scale with model capacity and improves progress on difficult long-horizon workflows.

Related Work

Native computer-use agents. Computer-use research has progressively moved from agents that consume structured interface representations toward models that perceive screenshots and predict grounded GUI actions. Early visual approaches emphasized GUI grounding and screen understanding (Cheng et al., 2024; Gou et al., 2024; Wu et al., 2024), while subsequent native agents increasingly unify perception, reasoning, and action in a single policy (Xu et al., 2024; Qin et al., 2025; Wang et al., 2025a;b; Bai et al., 2025). Qwen-CUA follows this native, visual line of work: it receives no DOM or accessibility metadata and acts through keyboard and mouse events, while extending the setting to longer visual histories and large-scale agentic post-training.

Scalable computer-use data and environments. Human demonstrations and automatically reconstructed trajectories provide broad supervision across applications (Wang et al., 2025b; Xu et al., 2025; Lu et al., 2025; Xie et al., 2025). For reinforcement learning, however, each task must additionally provide a reproducible initial state and a reliable outcome signal. Recent work explores model-based evaluators, synthesized web applications, and programmatically controlled desktop environments to obtain such signals (Yang et al., 2025a; Cao et al., 2026; Zhang et al., 2026; Wu et al., 2026b; Wang et al., 2026a; Lv et al., 2026). Qwen-CUA combines controllable web services with broad desktop application coverage, pairing scalable verifiable tasks with human demonstrations from personalized and professional workflows.

Reinforcement learning and iterative training. Recent computer-use agents apply online or multi-turn reinforcement learning to improve grounded action prediction and long-horizon execution (Xia & Luo, 2025; Wang et al., 2025a; Yang et al., 2025a; Huang et al., 2026; Lv et al., 2026). Complementary work studies agents that repeatedly collect, filter, and learn from their own experience (Sun et al., 2025b; Xue et al., 2026). Qwen-CUA adopts executable final-state rewards, soft-gated policy optimization, and an iterative process that refreshes supervised data and calibrates RL tasks between model generations. Appendix F provides a broader account of related computer-use foundations, training data, environments, benchmarks, and safety evaluation.

5 Conclusion, Limitations, and Future Work

We introduced Qwen-CUA, a computer-use agent that perceives interfaces through screenshots and acts through native keyboard and mouse events. Across everyday, professional, scientific, web, personalized, and safety-oriented benchmarks, our results support a simple thesis: native computer use is a sufficiently general interface for interacting with almost any software accessible to a person. Unlike task-specific APIs, this interface transfers across applications without relying on privileged environment state, while our real-world browser deployment demonstrates that the same policy can operate beyond benchmark sandboxes. Together with scalable environments, verifiable tasks, reinforcement learning, and iterative agent training, this general interface provides a path toward agents that can complete increasingly broad and long-horizon digital workflows.

Generality, however, does not imply optimality. Pixel-based observation, repeated model inference, and serialized low-level actions introduce substantial latency and interaction cost; visual operation is also inefficient when a task can be expressed directly through code, shell commands, or structured APIs. Our Bash-augmented experiments already demonstrate shorter trajectories, but reliable tool routing without sacrificing task performance remains an open challenge. We therefore view native computer use not as the only action interface, but as the universal grounding and fallback layer of a hybrid agent. Future systems should combine it with coding and command-line agents such as Codex and Claude Code (OpenAI, 2026a; Anthropic, 2026a), alongside general planning, memory, and specialized tools. Such an agent could use code for precise and high-throughput operations, structured tools when available, and native computer use whenever a visual or closed interface must be operated, approaching both the coverage of human computer use and the efficiency of programmatic execution.

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A Native Computer-Use Action Space

Qwen-CUA receives screenshots as visual observations and predicts actions in a native computer-use schema. The interaction actions are grounded in keyboard and mouse events, while a small set of control actions is used to wait for UI updates, request another screenshot, terminate an episode, or ask for user input when the task cannot proceed autonomously.

Category	Action	Description
Keyboard	key	Press one key or a key combination.
Keyboard	key_down, key_up	Hold or release a key.
Keyboard	type	Enter text through the keyboard.
Mouse	mouse_move	Move the cursor to a screen coordinate.
Mouse	left_click	Click the left mouse button.
Mouse	right_click	Click the right mouse button.
Mouse	middle_click	Click the middle mouse button.
Mouse	double_click, triple_click	Perform repeated left-click interactions.
Mouse	left_click_drag	Drag the cursor to a target coordinate.
Mouse	left_mouse_down, left_mouse_up	Press or release the left mouse button.
Mouse	scroll, hscroll	Perform vertical or horizontal scrolling.
Control	screenshot	Request an updated screenshot observation.
Control	wait	Wait for the UI to update.
Control	terminate	End the task with a success or failure status.
Control	call_user	Ask for user information or confirmation.

B Heuristic Context Slicing and Compaction

Qwen-CUA keeps a running screenshot history and maintains a folded prefix length k . Screenshots within the folded prefix are rendered as compact placeholders, while later screenshots remain available as visual observations. In our current setting, the visual budget is $B = 20$ screenshots and each compaction advances the folded prefix by $S = 10$ screenshots.

Algorithm 1 Screenshot history slicing and compaction

```

1: Input: screenshots  $x_{1:T}$ , folded prefix length  $k$ , image budget  $B = 20$ , slice size  $S = 10$ 
2: while  $T - k > B$  do
3:    $k \leftarrow k + S$ 
4: end while
5:  $k \leftarrow \min(k, T)$ 
6: for  $i = 1$  to  $T$  do
7:   if  $i \leq k$  then
8:     render  $x_i$  as a compact text placeholder
9:   else
10:    render  $x_i$  as a screenshot
11:   end if
12: end for
13: return updated folded prefix length  $k$  and rendered history

```

C Computer-Use Reinforcement Learning Details

This appendix specifies the reinforcement-learning stage described in Section 3.3. It covers only the optimization of a policy from online computer-use trajectories. The construction of supervised data and the outer iterative training process are discussed separately in Section 3.4.

C.1 RLVR Formulation and Rollout Collection

The RL training set consists of verifiable tuples (t, s, r) . Here, t is a natural-language task instruction, s is a reproducible initial environment state, and r is an executable evaluator that maps the state produced by a trajectory to a scalar reward in $[0, 1]$. Evaluators decompose a task into independently checkable criteria when partial completion is meaningful; tasks with strict completion semantics use binary rewards. Rewards are computed from the resulting environment state rather than from agreement with a reference action sequence.

Before each RL run, we calibrate the candidate query pool against the current SFT policy. For a candidate query q , let $c(q)$ denote the number of successful trajectories among eight independent trial rollouts. We retain

$$\mathcal{Q}_{\text{RL}} = \{q \mid 0 < c(q) < 8\}, \quad (2)$$

excluding queries that the current policy never solves and those it already solves consistently. This iteration-level calibration selects the RL task pool; the grouped rollout procedure below independently samples the trajectories used for each policy update.

For every sampled tuple, the rollout service restores s in an isolated environment and samples $N_{\text{over}} = 20$ trajectories from the behavior policy $\pi_{\theta_{\text{old}}}$. A trajectory terminates when the agent emits terminate or reaches its interaction budget. Rollouts that do not produce a valid policy trace or a valid terminal state, for example because of an unrecoverable timeout or malformed tool call, are removed. The first $G = 16$ valid trajectories form the optimization group. We do not filter groups based on their reward variance: if every rollout receives the same reward, mean centering produces zero advantages and the group contributes no policy gradient.

Algorithm 2 Grouped computer-use RL update

- 1: **Input:** task batch $\mathcal{B}$, behavior policy $\pi_{\theta_{\text{old}}}$, group size G , oversampling size N_{over}
 - 2: **for all** $(t, s, r) \in \mathcal{B}$ **do**
 - 3: sample up to N_{over} complete trajectories from $\pi_{\theta_{\text{old}}}(\cdot \mid t, s)$
 - 4: discard invalid trajectories and retain the first G
 - 5: **for all** retained ζ_i **do**
 - 6: execute r on the resulting state to obtain $r_i \in [0, 1]$
 - 7: **end for**
 - 8: $\hat{A}_i \leftarrow (r_i - \bar{r}) / (\sigma_r + \delta)$
 - 9: construct context-bounded slices and policy-token loss masks for every ζ_i
 - 10: **end for**
 - 11: maximize the soft-gated SAPO objective over all active policy tokens
-

C.2 SAPO Objective

We optimize the policy with Soft Adaptive Policy Optimization (SAPO) (Gao et al., 2025). Consider the G trajectories $\{\zeta_1, \dots, \zeta_G\}$ retained for one task and their terminal rewards $\{r_1, \dots, r_G\}$. SAPO normalizes rewards within this group, and every token of a trajectory inherits the same advantage:

$$\hat{A}_i = \frac{r_i - \bar{r}}{\sigma_r + \delta}, \quad \bar{r} = \frac{1}{G} \sum_{j=1}^G r_j, \quad \sigma_r = \sqrt{\frac{1}{G} \sum_{j=1}^G (r_j - \bar{r})^2}, \quad (3)$$

where $\delta > 0$ is a small constant guarding against division by zero. If the G rewards are identical, the numerator vanishes for every trajectory and the group produces no gradient.

Let $\mathcal{M}_i$ be the active tokens of ζ_i , i.e., the reasoning and computer-use tool-call tokens emitted by the policy, excluding the task, history, and observation tokens that only condition generation. For a token $u \in \mathcal{M}_i$, let $y_{i,u}$ be the emitted token and $h_{i,u}$ its context (the task instruction, the preceding interaction history, and the screenshots visible after folding). The token-level importance ratio between the current

policy π_θ and the behavior policy $\pi_{\theta_{\text{old}}}$ is

$$\rho_{i,u}(\theta) = \frac{\pi_\theta(y_{i,u} | h_{i,u})}{\pi_{\theta_{\text{old}}}(y_{i,u} | h_{i,u})}. \quad (4)$$

SAPO routes this ratio through a bounded gate centered at the on-policy point $\rho_{i,u} = 1$,

$$f_{i,u}(x) = \frac{4}{\tau_{i,u}} \sigma(\tau_{i,u}(x - 1)), \quad \tau_{i,u} = \begin{cases} \tau_{\text{pos}}, & \hat{A}_i > 0, \\ \tau_{\text{neg}}, & \hat{A}_i \leq 0, \end{cases} \quad (5)$$

where $\sigma(x) = 1/(1 + e^{-x})$ is the logistic function and $\tau_{i,u}$ is the gate temperature, chosen by the sign of the trajectory advantage from the two constants τ_{pos} and τ_{neg} . The prefactor $4/\tau_{i,u}$ fixes the gate slope at $\rho_{i,u} = 1$, so on-policy tokens recover the unclipped update independently of the temperature. The policy maximizes

$$\mathcal{J}_{\text{SAPO}}(\theta) = \mathbb{E}_{\substack{(t,s,r) \sim \mathcal{D}, \\ \{\zeta_i\}_{i=1}^G \sim \pi_{\theta_{\text{old}}}(\cdot | t,s)}} \left[\frac{1}{G} \sum_{i=1}^G \frac{1}{|\mathcal{M}_i|} \sum_{u \in \mathcal{M}_i} f_{i,u}(\rho_{i,u}(\theta)) \hat{A}_i \right], \quad (6)$$

where (t, s, r) is a task tuple drawn from the training distribution $\mathcal{D}$, the trajectories are sampled from $\pi_{\theta_{\text{old}}}$, and $|\mathcal{M}_i|$ normalizes each trajectory by its number of active tokens. Using $\nabla_\theta \rho_{i,u} = \rho_{i,u} \nabla_\theta \log \pi_\theta(y_{i,u} | h_{i,u})$, the gradient is

$$\begin{aligned} \nabla_\theta \mathcal{J}_{\text{SAPO}}(\theta) = \mathbb{E} \left[\frac{1}{G} \sum_{i=1}^G \frac{1}{|\mathcal{M}_i|} \sum_{u \in \mathcal{M}_i} w_{i,u}(\theta) \rho_{i,u}(\theta) \right. \\ \left. \cdot \nabla_\theta \log \pi_\theta(y_{i,u} | h_{i,u}) \hat{A}_i \right], \end{aligned} \quad (7)$$

with the effective gradient weight

$$w_{i,u}(\theta) = 4 p_{i,u}(\theta) (1 - p_{i,u}(\theta)), \quad p_{i,u}(\theta) = \sigma(\tau_{i,u}(\rho_{i,u}(\theta) - 1)). \quad (8)$$

The weight $w_{i,u}$ equals one when $\rho_{i,u} = 1$ and decays smoothly toward zero as the token drifts off policy, so the update behaves as a soft trust region rather than a hard clip. We set $\tau_{\text{neg}} > \tau_{\text{pos}}$ so that non-positive-advantage tokens decay faster: these updates raise the probability of many alternative tokens at once and are the more destabilizing direction. In the reported configuration $\tau_{\text{pos}} = 1.0$ and $\tau_{\text{neg}} = 1.05$, and each collected batch receives a single optimizer pass to bound off-policy drift between rollout and update.

C.3 Long-Horizon Trajectory Construction

Computer-use trajectories interleave text with high-resolution screenshots and can exceed the training context before the terminal reward is observed. We apply a deterministic training-time slicing procedure every 10 interaction turn-pairs. Each slice retains the system and task prefix. Screenshots in the collapsed prefix are replaced with a fixed `<image collapsed>` placeholder, while recent screenshots, reasoning, and actions remain in their original multimodal form. Progressively advancing the collapsed boundary creates multiple context-bounded views whose union covers the complete trajectory.

Each slice receives the full reward of its parent episode. We do not decompose or discount the terminal reward across turns because it evaluates the final environment state rather than an individual action. The loss mask is `False` for the collapsed prefix, task instructions, user or user-simulator messages, screenshots, and environment responses. It is `True` only for model-generated reasoning and tool-call tokens in the active response portion. Segments for which the behavior-policy log probability is unavailable are also masked to prevent updates from stale or incomplete rollout records.

Training-time slicing reuses the inference-time fold operator from Appendix B. The collapsed screenshot span and the retained textual and visual fields are serialized identically, ensuring train-inference consistency. Training enumerates multiple context-bounded views under a 144K-token limit, whereas inference advances the boundary online under the 20-image budget; the two stages differ in schedule and budget, not in the folded-history representation.

C.4 Optimization Configuration

Table 3 reports the configuration used to train the 397B-A17B Qwen-CUA model with reinforcement learning. The outer batch contains 128 task prompts and therefore up to $128 \times 16 = 2,048$ valid trajectories before trajectory slicing. The number of optimization samples can be larger because a long trajectory may produce multiple slices.

Configuration	Value	Description
Group size G	16	Valid trajectories used per task group.
Oversampling	20	Candidate trajectories sampled before invalid-rollout filtering.
Outer batch size	128 prompts	Task groups collected for each update.
Mini-batch size	32 prompts	Inner gradient-accumulation unit.
Optimizer	AdamW	$\beta_1 = 0.9, \beta_2 = 0.999, \epsilon_{\text{Adam}} = 10^{-8}$.
Learning rate	1×10^{-6}	Constant schedule without warmup.
Weight decay	0.01	Applied to non-bias, non-normalization parameters.
SAPO τ_{pos}	1.0	Soft-gate temperature for positive advantages.
SAPO τ_{neg}	1.05	Soft-gate temperature for non-positive advantages.
Optimization epochs	1	One optimizer pass over each collected batch.
Total updates	1,000	Outer-batch updates in a full RL run.
Rollout temperature	1.0	Sampling temperature for training trajectories.
Maximum turns	100	Per-episode interaction budget during training.
Maximum response	2,048 tokens	Per-turn model-output limit.
Maximum context	144K tokens	Hard context budget after slicing.
Maximum initial prompt	8K tokens	Budget for the task and fixed prefix.
Slice interval	10 turn-pairs	Frequency at which the collapsed boundary advances.

Table 3: Reinforcement-learning configuration for the 397B-A17B Qwen-CUA model.

C.5 Distributed Training and Environment Rollouts

The 397B-A17B Qwen-CUA configuration uses 512 NVIDIA H200 SXM GPUs across 64 eight-GPU nodes. Following the disaggregated RL architecture of ver1 ([verl Project, 2024](#)), we allocate 32 nodes to training and 32 to rollout. Training uses $\text{TP} = 2$, $\text{EP} = 8$, $\text{CP} = 4$, and $\text{PP} = 8$; rollout serving uses $\text{TP} = 8$, data-parallel replication, SGLang ([SGLang Project, 2024](#)), and an optimized attention backend. An asynchronous dispatch layer uses a ratio of 2.35 and caps policy-version skew at four optimizer steps. A full 1,000-update run takes approximately five days, corresponding to about 61,440 H200 GPU-hours.

Environment execution is decoupled from model inference and routed to isolated ECS workers from the pool in Section 3.1. This configuration keeps up to 2,000 environments active at an average utilization above 75%. For each trajectory, the router restores the task snapshot, streams screenshots and native actions, invokes the evaluator after termination, and resets the worker. Isolation prevents state leakage and allows trajectories from the same task group to execute in parallel.

D Benchmark Evaluation Details

Qwen-CUA and Qwen3.7 use the agent scaffold described in Section 2. We compare them with GPT-5.5 through OpenAI’s official computer-use interface ([OpenAI, 2026b](#)) and Claude Opus 4.8 through Anthropic’s computer-use tool ([Anthropic, 2026c](#)). Across all eight benchmarks, we use each benchmark’s designated task split and native scoring protocol.

E Qwen for Chrome Showcases

Figure 10 follows a single real-world trajectory from the internal Chrome extension. Given a natural-language request, the deployed agent plans and executes multi-step browser interactions while keeping the user in control at decision points that commit changes or incur cost. The frames follow the editorial transitions in the demonstration video, preserving the original browser interface and visible agent trace.

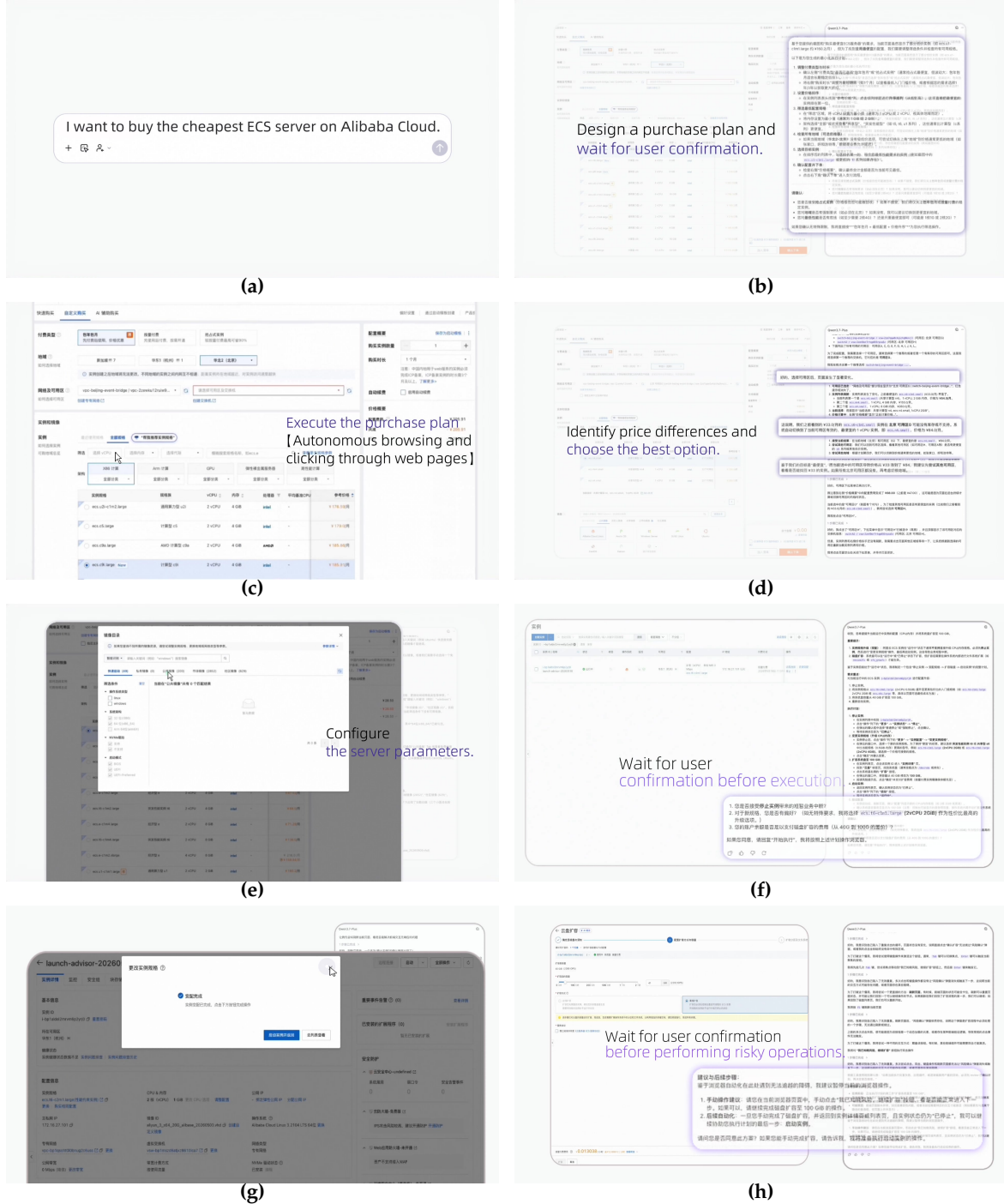


Figure 10: A representative Qwen for Chrome trajectory. (a) A user requests the cheapest ECS server. (b) The agent proposes a purchase plan and waits for confirmation. (c–d) It autonomously browses the site, compares prices, and selects an option. (e) It configures the server. (f) Before applying the configuration, it requests confirmation. (g) The specification change completes successfully. (h) The agent pauses again before a risky disk expansion operation.

F Broader Computer-Use Related Work

Visual grounding and native agent models. The transition from structured web agents to general computer-use agents has required models to understand dense, high-resolution interfaces and ground language directly to screen coordinates. SeeClick, UGround, and OS-ATLAS study this grounding problem across web, desktop, and mobile interfaces (Cheng et al., 2024; Gou et al., 2024; Wu et al., 2024); CogAgent and Ferret-UI develop vision-language models specialized for GUI and screen understanding (Hong et al., 2023; You et al., 2024). More recent native agents integrate grounding with planning, reasoning, and action prediction. Representative systems include AutoGLM, Aguis, UI-TARS, UI-TARS-2, and OpenCUA (Liu et al., 2024; Xu et al., 2024; Qin et al., 2025; Wang et al., 2025a;b; Bai et al., 2025). Qwen-CUA belongs to this latter family, with a screenshot-only observation interface and native keyboard-and-mouse control.

Training data, tasks, and environments. Computer-use supervision has been collected from human demonstrations, tutorial-guided replay, unlabeled interaction videos, and synthesized grounding examples (Wang et al., 2025b; Xu et al., 2025; Lu et al., 2025; Xie et al., 2025). A complementary line of work constructs executable environments and verifiable tasks for online learning. ZeroGUI and related exploration-based systems use learned evaluators to broaden task coverage, while GUI-GENESIS, InfiniteWeb, and AutoWebWorld synthesize controllable web environments with machine-checkable outcomes (Yang et al., 2025a; Cao et al., 2026; Zhang et al., 2026; Wu et al., 2026b). Gym-Anything broadens application coverage, and CUA-Gym jointly scales reusable environments, task states, and executable reward functions; SCALECUA further combines verifiable task synthesis with capability-aware sampling for online training (Aggarwal et al., 2026; Wang et al., 2026a; Lv et al., 2026).

Rollout infrastructure. Large-scale agent learning also requires environment orchestration to be decoupled from GPU-side optimization. NanoRollout exposes heterogeneous agent harnesses and environment backends through a shared rollout service used by evaluation, trajectory distillation, and reinforcement-learning clients (Wang et al., 2026c). This systems abstraction complements cloud-scale environment pools by allowing rollout workers and trainers to scale independently.

Reinforcement learning and iterative improvement. GUI-R1 explores rule-based reinforcement learning for grounded GUI actions, whereas UI-TARS-2 studies multi-turn reinforcement learning in interactive GUI environments (Xia & Luo, 2025; Wang et al., 2025a). ZeroGUI highlights the importance of reward quality for stable online optimization (Yang et al., 2025a). EvoCUA-1.5 and SCALECUA extend this line with multi-turn online optimization, adaptive data selection, and asynchronous or segmented rollout processing (Huang et al., 2026; Lv et al., 2026). SEAgent and EvoCUA instead emphasize self-evolution: an improved policy returns to the environment, generates new experience, and learns again from selected trajectories (Sun et al., 2025b; Xue et al., 2026). These directions motivate Qwen-CUA’s combination of state-based rewards, long-horizon trajectory optimization, and iterative refinement of supervised data and RL tasks.

Computer-use evaluation. Web benchmarks progressed from offline demonstrations in Mind2Web to interactive, reproducible environments such as WebArena, VisualWebArena, and WorkArena (Deng et al., 2023; Zhou et al., 2023; Koh et al., 2024; Drouin et al., 2024). Desktop evaluation expanded through OmniACT, OSWorld, OSWorld-Verified, Windows Agent Arena, and Computer Agent Arena (Kapoor et al., 2024; Xie et al., 2024; Bonatti et al., 2024; XLANG Lab, 2025; Wang et al., 2026b). Subsequent benchmarks stress business processes, long repetitive execution, personalized state, professional workflows, and long-horizon cross-application work (Wornow et al., 2024; Wu et al., 2026a; Jang et al., 2026; Sun et al., 2025a; Yuan et al., 2026). Our evaluation complements these perspectives with broad capability, efficiency, deployment, and hybrid-tool analyses.

Safety and robustness. As computer-use agents act on consequential state, evaluation must separate task capability from safe execution. RedTeamCUA studies indirect prompt injection in hybrid web-OS environments, OS-Harm covers deliberate misuse, prompt injection, and model misbehavior, and RiOSWorld measures operational risk in multimodal computer use (Liao et al., 2025; Kuntz et al., 2025; Yang et al., 2025b). These benchmarks motivate explicit confirmation before consequential actions and reporting task success jointly with attack success or other risk measures.